Real course details

Same layout as the first course. The first hallway, inset, bins, and second hallway are all the same.

Left-side inset -> bottom edge of first pair of bins: 41”
Right wall -> top right edge of first pair of bins: 57”
Front edge of first pair of bins -> back edge of second pair of bins: 40”
Left wall -> top left edge of second pair of bins: 57.5”
Front right wall corner -> front right edge of third bin: 58”
Front left edge of third bin -> left wall: 36”

Bag command

ros2 bag record -o real_bag1 /diffbot_base_controller/cmd_vel /cmd_vel_raw /diffbot_base_controller/odom /odometry/filtered/local /odometry/filtered/global /global_costmap/costmap /local_costmap/costmap /velodyne_points_nav /imu/data /plan /odom_goal_pose /odom_goal_pose_rviz /tf /dynamic_joint_states /tf_static /local_costmap/published_footprint /global_costmap/published_footprint /rosout /behavior_tree_log /speed_limit /speed_governor_info /local_costmap/costmap_raw /global_costmap/costmap_raw /velodyne_points

Run #1 (Teleoperation, all changes as-is)

bag name: real_bag1
This bag shows the manual driving of the course via teleoperation to prove it can be completed comfortably.

The robot’s lidar is high up, and the trashcans (and real obstacles in the course) are expected to fall out of the lidar’s view as the robot drives close to them, since the Velodyne VLP-16 only has a vertical FOV of 30 degrees, which means 15 degrees downward. This should NOT cause the occupied costmap cells to disappear; that would be disastrous. That is what was observed here, and this must be fixed before moving on. Anything in the costmap that isn’t in direct view of the sensor should not be cleared. The manual drive was stopped prematurely after we observed the obstacle disappearing in the map.

Run #2 (Teleoperation, all changes as-is)

bag name: real_bag2
Same thing as #1, but driven further along the course. Observed all the obstacles disappearing as the robot neared them.

Notes

This is what we observed happening with obstacle_layer, and was also the reason why clearing was turned off. You may want to investigate the svtl layer for this - maybe it has more advanced clearing capabilities (Kilted SVTL layer is already installed on this machine). Regardless, this issue must be fixed. Don’t change anything else; just try to fix this problem. The change report can be shortened accordingly.